

A Guide to the Data Handling Web Application

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This document aims to provide everything a user must know to understand my web application. Through this text, a user should be able to correctly handle their data on the application, and recognize reasons why they maybe have been unable to do so. The figures on this document have very low quality, so i highly recommend zooming into them while reading through the text.

File Format

The web application expects the data file to be formatted in a specific way. Figure 1 presents an example of how the file could look.

```
Time;Name_A;Name_A;Name_B;Name_C;Name_D;Name_E;Name_F;Name_G;Name_H;Name_I;Name_J;Name_K;Name_L;Name_M;Name_N;Name_O
2016-06-02 00:21:59.871;65533;65533;65532;65534;0;0;65535;0;0;2;0;65535;0;0;0;0
2016-06-02 00:22:00.890;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:01.906;65533;65533;65533;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:02.922;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:03.931;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:04.940;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:05.945;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:06.957;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:07.969;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:08.980;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;1;0;0
2016-06-02 00:22:09.986;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:10.993;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:12.000;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:13.006;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:14.017;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:15.029;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:16.036;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:17.040;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:18.054;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:19.065;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:20.072;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:21.085;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
2016-06-02 00:22:22.096;65533;65533;65532;65534;0;0;65535;0;0;0;65535;0;0;0;0
```

Figure 1: An example of a data file

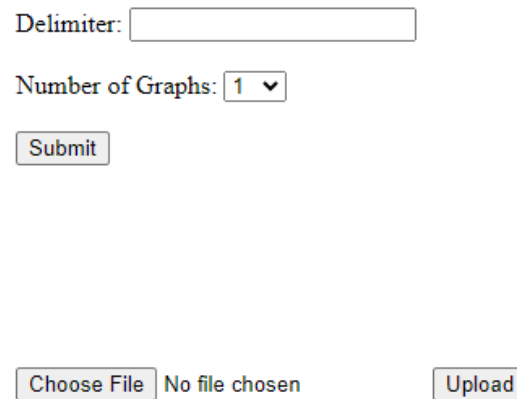
In this case the data file is a text file, which is not an obligation. In addition, the delimiter is a semicolon, but any delimiter suffices.

The headers must be on the top row and have unique names. Additionally, if there is date-time formatted representation of time, it must be as YEAR-MONTH-DAY HOURS:MINUTES:SECONDS.MILLISECONDS. The program will attempts to interpret date-time values which are written differently than stated, but there is no guarantee that it will be interpreted correctly or interpretable at all.

It is important that the data file itself is a plain text file format and not a binary file format. For example, if the user wishes to analyze a .xlsx file, it must be saved as a .csv file before uploaded to this web application.

The First Page

Figure 2 demonstrates how my web application looks when initially opened.



Delimiter:

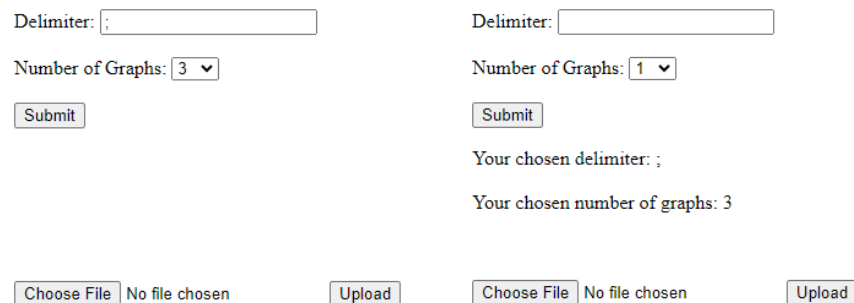
Number of Graphs: ▼

No file chosen

Figure 2: The first page of the web application

As can be seen, the page contains a text input for the data's delimiter, a drop-down list of the number of graphs one wishes to be presented, a submit button, and finally a field to upload the data file.

Submitting the delimiter and number of graphs is the first step. Figure 3 illustrates how the website should look before (left) and after (right) submitting.



Left (Before Submitting):

Delimiter:

Number of Graphs: ▼

Right (After Submitting):

Delimiter:

Number of Graphs: ▼

Your chosen delimiter: ;

Your chosen number of graphs: 3

No file chosen No file chosen

Figure 3: The website before and after submitting inputs

As is noticeable, the web page will reflect which delimiter and how many graphs were chosen by the user. If no delimiter and number of graphs is submitted, the program is set to choose ',' as the delimiter and make one graph.

The data file can now be chosen and uploaded, leading to the second page.

The Second Page (/upload)

Figure 4 presents how the second page of my web application might look.

Was unable to turn cell (1 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (2 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (3 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (4 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (5 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (6 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (7 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (8 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (9 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (10 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
Was unable to turn cell (11 down, 78 right) to a number. This value will be ignored in plotting. Cell value:
More cells will be ignored in plotting. These 10 above are examples of which values

File is done being read

Graph nr. 1
Choose your x-axis:
Choose your y-axis:

Graph nr. 2
Choose your x-axis:
Choose your y-axis:

Graph nr. 3
Choose your x-axis:
Choose your y-axis:

Figure 4: The second page of the web application

The page begins by informing the user which values will further be ignored due to their inability to be represented as a number. The position of these values in the data file, along with the values themselves, are explicitly stated. Figure 5 illustrates how the position of the values are calculated for, in this case, a text file where the delimiter is a semicolon.

Time	Name_A	Name_B	Name_C	Name_D	Name_E	Name_F	Name_G	Name_H	Name_I	Name_J	Name_K	Name_L	Name_M	Name_N	Name_O
2016-06-02 00:21:59.871	65533	65533	65532	65534	0	0	65535	0	0	2	0	65535	0	0	0
2016-06-02 00:22:00.890	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:01.906	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:02.922	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:03.931	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:04.940	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:05.945	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:06.957	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:07.969	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:08.980	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	1	0
2016-06-02 00:22:09.986	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:10.993	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:12.000	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:13.006	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:14.017	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:15.029	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:16.036	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:17.040	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:18.054	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:19.065	65533	65534	65533	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:20.072	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:21.085	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0
2016-06-02 00:22:22.096	65533	65533	65532	65534	0	0	65535	0	0	0	0	65535	0	0	0

CELL (1 DOWN, 1 RIGHT)
CELL (4 DOWN, 3 RIGHT)
CELL (11 DOWN, 6 RIGHT)

Figure 5: How the position of a value is calculated

It is thereby simple to realize that the data file uploaded on the first page, leading to the screen on Figure 4, had an empty 78th column. This is because the messages shows that every value ignored is 78th in every row, and the value itself is presented as blank.

It is notable that these messages will only show to the user if there are values in the data file that the program cannot interpret.

Further, Figure 4 shows that the second page continues. Graph numbers along with dropdown lists of the headers are given to choose an x-axis and y-axis per graph. The number of graphs presented are decided by the number of graphs the user chose on the first page. The user should now choose which columns of data they wish to see the relation of for each graph, and finally submit.

It is of importance that the user does not submit a graph where date-time formatted values are on the y-axis, as the program will not understand how to plot such a graph.

The Third Page, The Fourth Page (/plot, /soc)

Figure 6 demonstrates how the third page of my web application could look if, on the previous page, the user picks 'Time' for all the x-axes and 'Temp_Leg2_Batt_FWD', 'AM_SOC_FWD', and 'AM_Voltage_FWD' respectively for the y-axes.



Figure 6: The third page of the web application

In the top right corner of each graph, the user should notice that there are tools that can be used to download the plot as a png, zoom in and out of the plot, and pan around the plot.

Larger numbers in the data file will be much lower numbers on the plots. This is due to the program's ability to handle 16-bit values where it has been ignored that the first bit is a sign bit. Simply, this means any number over 32767 is interpreted to be 65536 lower than that number.

Under each graph, there is an option to analyze the spikes of the plot. To analyze the spikes, the program needs a threshold to decide the minimum change in the y-axis that registers as a spike. Figure 7 provides an example of the output, if the user wished to analyze the spikes of 'AM_SOC_FWD on Time' on Figure 6. The threshold is set to 5.

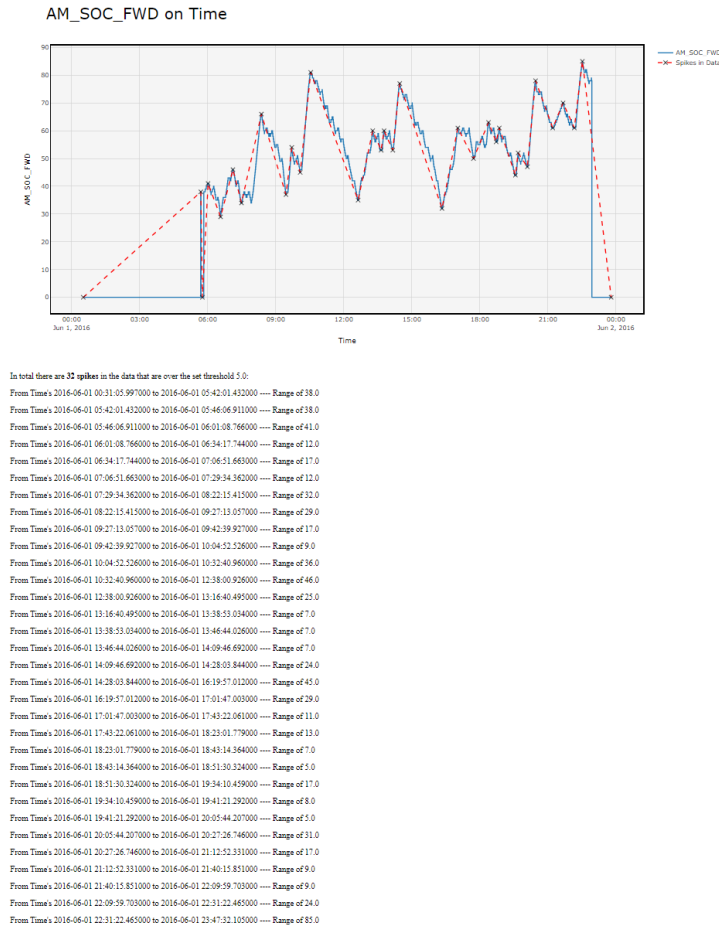


Figure 7: The fourth page of the web application

As can be seen, a graph is provided with all the spikes in the data that are over the set threshold. Additionally, messages are sent to the user informing how many spikes there are and where the spikes are located on the plot.

Final Words

I hope this document was an understandable and helpful explanation to how my data handling web application works. I wish to thank Wärtsilä Norway for giving me the opportunity to build and expand this application.